Total No. of Questions : 4]

SEAT No. :

# PD123

[Total No. of Pages : 1

## [6410]-445
## T.E. (E & TC) (Insem)
## CELLULAR NETWORKS
## (2019 Pattern) (Semester - II) (304192)

*Time : 1 Hour]* *[Max. Marks : 30*

***Instructions to the candidates:***

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of Calculator is allowed.***
5) ***Assume suitable data if necessary.***

***Q1)*** a) Explain free space propagation model & derive the formula $P_t = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2}$ **[6]**

b) Discuss factors influencing Small scale fading. **[4]**

c) What is diversity? Explain its types in brief. **[5]**

OR

***Q2)*** a) Explain ground- reflection scenario. **[5]**

b) Employing the Hata model, compute the median loss at a distance d = 8 km, when the carrier frequency $f_c$= 2.1 GHz, $h_{te}$ = 40 m, $h_{re}$= 2 m for a large city. **[4]**

c) Explain in details concept of channel estimation in wireless communication. **[6]**

***Q3)*** a) Explain with neat diagram working of OFDM transmitter. **[6]**

b) Define smart antenna and explain the types of smart antenna in brief. **[5]**

c) Consider IHI$^2$ and data power P = 10 dB with noise power $\sigma_n^2$ = 0 dB. Derive the SNR/SINR with and without a carrier frequency offset of $\varepsilon$ = 5 % in a WiMAX system. **[4]**

OR

***Q4)*** a) Explain multicarrier transmission. **[5]**

b) Write advantages, disadvantages & applications of MIMO-OFDM. **[6]**

c) Write note on MIMO. **[4]**